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Grouped with:

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ENGG 151.01-A

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Early Work for Project 3: Linear Time-Invariant System Simulator

```
PS D:\Peter's Stuff\New Coding\Estacio_Peter\ENGG 151\Project 3\v4> ./main
Type help for more information
>help
Available commands:
  system <filename>  : Load LTI system coefficients from file.
  signal <filename>  : Process a signal file through the current system.
  <number>           : Input a floating-point sample into the system.
  clear              : Reset simulation history to 0.0.
  cls                : Clear the screen.
  exit               : Terminate the application.
>a
Invalid command. Type 'help' for info.
>0
ERROR: No LTI system has been defined yet.
>1p
Invalid command. Type 'help' for info.
>system test.txt
Invalid system file read, M+1 and N detected as 0.
ERROR: Could not obtain LTI system from test.txt
>system lab5.system
System obtained from lab5.system. recursive coefs: 2, nonrecursive coefs: 3
>0.0000000000000000
0
>0.0980171403295606
0.0322507
>0.1950903220161280
0.0797881
>signal test.txt
Signal file 'test.txt' with starting index 0 and duration 3 read.
0.290285        0.141681
0.382683        0.215166
0.471397        0.29668
>clear
Simulation restarted. Initial conditions cleared.
>exit
PS D:\Peter's Stuff\New Coding\Estacio_Peter\ENGG 151\Project 3\v4>
```

screenshot – executable main.exe run with run basic test and additional input validation tests

```
ENGG 151 > Project 3 > v4 > lab5.system
3 nNonRecursiveCoefs = M+1 = 3; M = 2
2 nRecursiveCoefs = N = 2
0.3290316064918401 b0 = 0.3290316064918401
-0.31460820551056307 b1 = -0.31460820551056307
0.11124319871419965 b2 = 0.11124319871419965
-1.439788397628722 a1 = -1.439788397628722
0.5654549973241987 a2 = 0.5654549973241987
from DSP Lab Activity: Magnitude Response of a Biquad Filter
```

```
ENGG 151 > Project 3 > v4 > test.txt
0.2902846772544620
0.3826834323650900
0.4713967368259980
```

screenshots – lab5.system file and test.txt signal file used for basic test

```
ENGG 151 > Project 3 > v4 > ltisim-log.txt
new system
3
2
0.329032
-0.314608
0.111243
-1.43979
0.565455
ready
0   0
0.0980171   0.0322507
0.19509   0.0797881
0.290285  0.141681
0.382683  0.215166
0.471397  0.29668
clear
```

screenshot – ltisim.log.txt file output after basic test

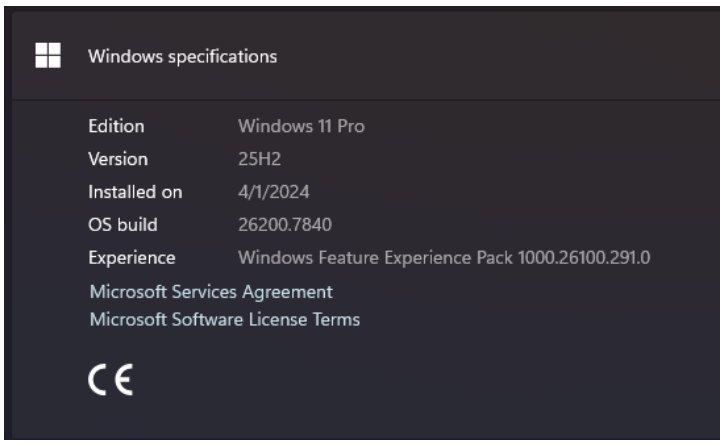
```
Starting build...
cmd /c chcp 65001>nul && C:\msys64\mingw64\bin\g++.exe -fdiagnostics-color=always -g *.cpp -o "D:\Peter's Stuff\New Coding\Estacio_Peter\ENGG 151\Project 3\v4\main.exe"

Build finished successfully.
 * Terminal will be reused by tasks, press any key to close it.
```

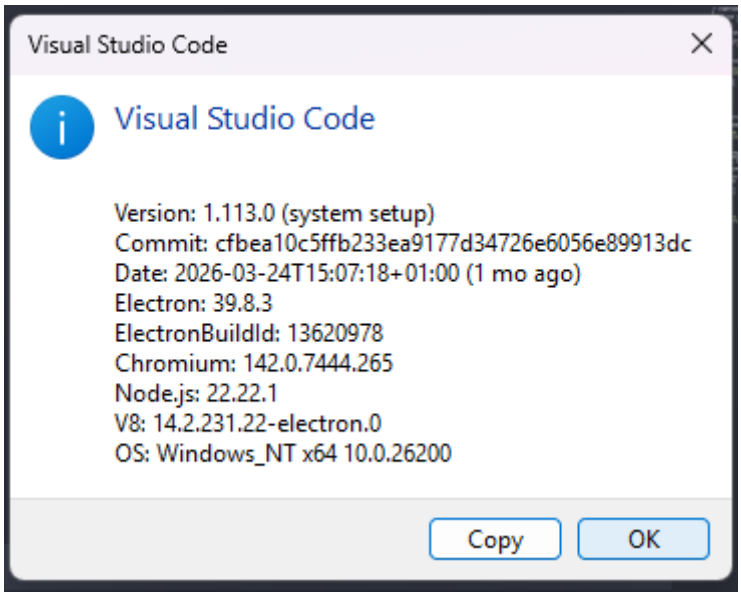
screenshot – compilation on Visual Studio Code

```
PS D:\Peter's Stuff\New Coding\Estacio_Peter> g++ -v
Using built-in specs.
COLLECT_GCC=C:\msys64\mingw64\bin\g++.exe
COLLECT_LTO_WRAPPER=C:/msys64/mingw64/bin/../lib/gcc/x86_64-w64-mingw32/15.2.0/lto-wrapper.exe
Target: x86_64-w64-mingw32
Configured with: ../gcc-15.2.0/configure --prefix=/mingw64 --with-local-prefix=/mingw64/local --with-native-system-header-dir=/mingw64/include --libexecd
ir=/mingw64/lib --enable-bootstrap --enable-checking=release --with-arch=nocona --with-tune=generic --enable-mingw-wildcard --enable-languages=c,lto,c++,
fortran,ada,objc,obj-c++,jit --enable-shared --enable-static --enable-libatomic --enable-threads=posix --enable-graphite --enable-fully-dynamic-string --
enable-libstdcxx-backtrace=yes --enable-libstdcxx-filesystem-ts --enable-libstdcxx-time --disable-libstdcxx-pch --enable-lto --enable-libgomp --disable-l
ibssp --disable-multilib --disable-rpath --disable-win32-registry --disable-nls --disable-werror --disable-symvers --with-libiconv --with-system-zlib --w
ith-gmp=/mingw64 --with-mpfr=/mingw64 --with-mpc=/mingw64 --with-isl=/mingw64 --with-pkgversion='Rev8, Built by MSYS2 project' --with-bugurl=https://gith
ub.com/msys2/MINGW-packages/issues --with-gnu-as --with-gnu-ld --with-libstdcxx-zoneinfo=yes --disable-libstdcxx-debug --enable-plugin --with-boot-ldflag
s=-static-libstdc++ --with-stage1-ldflags=-static-libstdc++
Thread model: posix
Supported LTO compression algorithms: zlib zstd
gcc version 15.2.0 (Rev8, Built by MSYS2 project)
PS D:\Peter's Stuff\New Coding\Estacio_Peter>
```

compiler version – gcc version 15.2.0 (Rev8, Built by MSYS2 project)



operating system - Windows 11 Pro, 25H2, build 26200.7840



IDE – Visual Studio Code ver 1.113.0

Code locator:

code items	file name	line numbers
separate processing of first line of signal file	ltisim.cpp	78-102
input validation of integers in processing signal files and system files	ltisim.cpp	84-92, 155-165
input validation of floating point numbers in processing signal files and system files	ltisim.cpp	87-99, 109-113, 175-196
allocation of fixed-size arrays for simulating LTI systems	main.cpp	17-18, 140-143
simulation function(s) (that compute the output(s) given input(s) to the system)	ltisim.cpp	213-246

Project Evaluation		
Item	Points	Rubrics
early work	8/8	(all or nothing) 8: at most 10% of the code in the final implementation differs from that in early work submission
implementation and testing environment report	4/4	4 - all instructions followed correctly
Basic Test	50/50	0 - at least one computation in the basic test is not done correctly or C++ vectors or equivalent structures are used in simulating an LTI system 50 - correctly simulates the LTI system provided in the basic test for all inputs provided
	6/6	6 - screenshot submitted showing the correctly computed output as it appears on the console output
generality of simulation	6/6	6 - LTI systems are simulated correctly and efficiently at all times
simulation function(s)	6/6	6 - LTI systems are simulated correctly and efficiently at all times, by function(s) as specified
processing of signal files	4/4	4 - The file is opened only once and the first line of the signal file is processed correctly and separately from code processing the rest of the signal file
processing of system files	4/4	4 - The file is opened only once and the first values of the file are used as bases for allocating a fixed amount of memory for simulating the system
input validation	4/4	4 - integers and floating point numbers are fully validated when signal and system files are processed
quality of application, user interface and program flow	4/4	areas to check (4): robustness of application, correct program flow, appropriate feedback and outputs, proper support of all required commands 0 - application sometimes behaves erratically (not robust), or deficiencies in two other areas 2 - robust application, deficiencies in at most one other area 4 - excellent in all areas
self-evaluation	4/4	4 - self-evaluation accurate (or evaluating this item leads to an error)
total	100/100	